Drug Design

How a designed molecule becomes a useful candidate

Small molecules, protein binders and antibodies

The design brief

Desired effect

Change a disease-relevant biological process.

Target access

Reach the relevant tissue and molecular site.

Selectivity

Separate the intended effect from harmful interactions.

Developability

Support a stable, manufacturable and deliverable molecule.

Three molecular design routes

Route	Design object	Typical starting point	Early constraint
Small molecule	Chemical scaffold and substitutions	A pocket or ligand-binding mechanism	Chemistry, selectivity and exposure
Peptide / minibinder	Sequence, fold and interface	A molecular surface or interaction	Stability and delivery
Antibody	Framework, CDRs and format	An accessible epitope	Developability and format

VHH and scFv are antibody-derived formats, distinct from non-antibody minibinders.

Small-molecule design

A chemical series connects design to measurement.

Synthesis and biological testing close the loop.

Explore chemistry

RDKit and REINVENT4

Propose binding poses

AutoDock Vina or AlphaFold 3

Prioritize affinity

Boltz-2 or alchemical calculations with OpenFE

Protein and minibinder design

Joint generation

RFD4-Proteina
sequence + structure

Backbone and sequence design

RFdiffusion → **ProteinMPNN**

→ **Complex prediction and candidate selection**

BindCraft integrates design and filtering in an experimentally evaluated binder workflow.

Antibody design

Framework

A structural context for the binding loops.

→

CDRs

Binding geometry at a chosen epitope.

→

Format

VHH, scFv, Fab or full antibody.

RFantibody and RFD4-Proteina

Epitope-directed design or CDR design.

TAP

Calculated developability flags.

A tool map by decision

Decision	Representative tools	Main output
What to propose?	REINVENT4, RFD4-Proteina, RFdiffusion, BindCraft	Candidate molecules
Which sequence?	ProteinMPNN	Designed sequences
Which complex?	AlphaFold 3, Boltz-2, Vina	Poses and assessment
Which affinity change?	Boltz-2, OpenFE	Context-dependent estimates
Which liabilities?	RDKit descriptors, TAP	Property flags

Tool roles overlap. Each output supports a different decision.

What the evidence establishes

Readout	What it can support	What still needs evidence
pLDDT, PAE, ipTM / ipSAE	Confidence in a predicted fold or interface	Affinity and biological effect
Binding and kinetics	Target engagement under the assay conditions	Desired biological function
Functional activity	A defined biological response	Exposure and tolerability
Stability, exposure and safety data	Feasibility of further development	Benefit in the intended setting

The strength of the conclusion must follow the kind of evidence.

Candidate selection

Biology

Mechanism and function

Molecular evidence

Binding and structural plausibility

Feasibility

Stability, exposure and production

Which result would change the decision?

Keep evidence types inspectable.
Retain useful molecular diversity.
Resolve the largest uncertainty next.

A useful first study

A reasoned choice of which molecule to develop.

01 State the mechanism and required properties.

02 Choose a design route and a clear baseline.

03 Build a diverse shortlist. Combine evidence.

04 Measure binding. Test function. Iterate.